



SCIENTIFIC REPOSITORY RECORD

PHAGE PROFILE — P002

GTU-P002 · GSPR-0002 · Pseudomonas aeruginosa

Verified

Lytic

Linear dsDNA

Public

Version 1.0

1. PHAGE IDENTITY



Repository ID GTU-P002
Phage name P002
Alternative name Not recorded
Repository accession GSPR-0002
Current status Verified

2. ISOLATION INFORMATION



Isolation date 22 Sept 2024
Isolation location Pirana Pumping Yard, Ahmedabad, Gujarat, India
GPS coordinates 23.0305, 72.58
Sample type Sewage
Environmental source Municipal wastewater
Collected by Ashutosh Gupta
Isolation method Enrichment method
Host used for isolation Pseudomonas aeruginosa PAO1

3. HOST INFORMATION



Primary host Pseudomonas aeruginosa
Host strain PAO1
ATCC number ATCC 27853
Host origin Clinical isolate
Gram stain Gram-negative
Clinical importance ESKAPE pathogen
Antibiotic resistance profile MDR

4. TAXONOMIC CLASSIFICATION



Order Caudovirales
Family Myoviridae
Subfamily Not recorded
Genus Not recorded
Species Not recorded
ICTV status Pending / Classified
Genome type Linear dsDNA

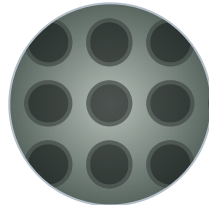
5. MORPHOLOGICAL CHARACTERISTICS



Capsid shape Icosahedral

Tail type Contractile
Tail length 152 nm
Collar Present
Tail fibers Present
Capsid diameter 144 nm
Tail width 18 nm
Base plate Present
MorphotypeA1

6. PLAQUE CHARACTERISTICS



Plaque diameter 2 mm

Plaque appearance Clear
Margin Smooth
Halo Present
Elevation Flat

Clear circular plaques, approx. 2 mm, smooth margins with halo.

7. BIOLOGICAL CHARACTERISTICS



Adsorption time 20 minutes
Adsorption rate constant 1.48×10^{-11} ml/min
Latent period 30 minutes
Eclipse period 20 minutes
Rise period 50 minutes
Burst size 18 PFU/infected cell
Optimal MOI 0.01
Life cycle Strictly lytic

8. GROWTH PARAMETERS



Optimal growth temperature 37°C
Optimal pH 7
Optimal host density $OD_{600} = 0.22$
Maximum PFU 2.86×10^{10} PFU/ml

ONE-STEP GROWTH CURVE



9. HOST RANGE



Bacterial host	Strain	Result
Pseudomonas aeruginosa	PAO1	✓ Susceptible
Pseudomonas aeruginosa	PA21848	✓ Susceptible
Pseudomonas aeruginosa	PAS101	✗ Resistant / no lysis
Klebsiella pneumoniae	KP01	✗ Resistant / no lysis
Escherichia coli	EC01	✗ Resistant / no lysis
Acinetobacter baumannii	AB5075	✓ Susceptible

6
hosts tested

3
susceptible

50%
host range

10. TEMPERATURE STABILITY	
Temperature (°C)	Stability
-80	Stable
-20	Stable
4	Stable
25	Stable
37	Stable
45	Stable
55	Stable
65	Stable
75	Stable
85	Reduced
95	Inactive
100	Inactive

11. pH STABILITY	
pH	Stability
1	No
2	No
3	No
4	Low
5	Yes
6	Yes
7	Excellent
8	Excellent
9	Good
10	Moderate
11	Low
12	No
13	No
14	No

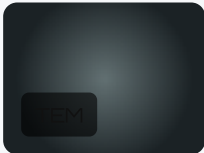
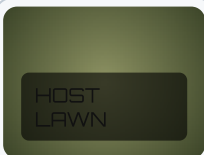
12. GENOME INFORMATION	
Sequencing platform	Oxford Nanopore
Genome size	163,245 bp
GC content	56.2%
Genome type	Linear dsDNA
Genome ends	Terminal repeat
Assembly	Complete
Coverage	120×
GenBank accession	ON123456

13. GENOME ANNOTATION	
Total ORFs	231
Coding density	95%
tRNA genes	2
rRNA	None
Integrase	Absent
Repressor protein	Absent
Virulence genes	Not detected
AMR genes	Not detected
Lysogeny genes	Absent
Lifestyle prediction	Lytic

14. FUNCTIONAL PROTEINS	
Protein	Predicted function
Major capsid protein	Structural
Portal protein	DNA packaging
Terminase large	Packaging
Tail fiber	Host recognition
Tail sheath	Infection
Endolysin	Cell lysis
Holin	Cell lysis
Spanin	Cell lysis
DNA polymerase	Replication

15. LABORATORY INFORMATION	
Current stock titer	2.8 × 10 ¹⁰ PFU/ml
Storage buffer	SM buffer
Storage temperature	-80°C
Cryoprotectant	20% glycerol
Freeze-thaw cycles	< 3
Freezer location	GTU Freezer 2
Rack	B
Box	7
Position	D4

16. EXPERIMENTAL PROTOCOLS (SOPs)		
 Isolation Protocol PDF	 Propagation Protocol PDF	 Purification Protocol PDF
 DNA Extraction PDF	 Plaque Assay PDF	 Spot Assay PDF
 TEM Preparation PDF		

17. IMAGES	
 TEM image	 Plaque image
 Host lawn	 Environmental sample

18. DOWNLOADS		
 Metadata JSON Download	 Metadata CSV Download	 Genome FASTA Download
 Annotation (GBK) Awaiting file	 Proteins (FAA) Awaiting file	 Raw Data (FASTQ) Awaiting file
 PDF Report Awaiting file		

19. APPLICATIONS	
Phage therapy	Biofilm removal
Food safety	Agriculture
Wastewater treatment	
Phage cocktail candidate	

20. PUBLICATIONS	
	Gupta A. et al. (2024). Isolation and characterization of novel lytic phage P002 against Pseudomonas aeruginosa. (In preparation)
	GSBTM Project Report 2024
	Conference Poster — Phage Repository Update 2024

21. REPOSITORY METADATA	
Date added	01 Oct 2024
Last updated	15 May 2025
Verified by	Dr. XYZ
Curator	GTU Phage Team
Data version	1.0
Quality status	Verified
Availability	Public